
IAL Physics Unit 6

Rapid Revision

WPH16 — Practical Skills in Physics II

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* guide
(18 past papers and mark schemes, October 2020 - June 2026)

50 marks · 1 h 20 min · always exactly four questions
A mark is worth about **1.5 minutes**

Roughly 20 of the 50 marks each session come from uncertainty arithmetic and graph plotting alone — both pure technique, no physics recall. Secure those first.

TRIAGE — WHAT SITS IN EACH SLOT

Slot	Marks	Almost always	Sometimes
Q1	6-13	Short measurement skills: resolution, how to measure something accurately, a small calculation, criticise recorded data	A sketch graph, control variables, a "show that"
Q2	6-13	Either a 6-mark "devise a plan" or a cluster of technique / safety / improvement marks	A circuit diagram, a data-logger evaluation
Q3	13-22	The graph question. Process data into logs or ln, plot, take a gradient, decide whether a proposed relationship holds	A second constant from the intercept; half-life from a decay graph
Q4	12-19	The uncertainty question. Instrument choice, mean $\pm$ uncertainty of repeats, propagate through a formula, then "deduce whether the material is X"	A "show that the uncertainty is about..." step

Skill	Papers	Sheet
Uncertainty propagated through a formula	18/18	3
Plot a graph on the printed grid (6 marks)	18/18	4
State / explain a technique to improve a measurement	17/18	2
Percentage uncertainty	16/18	3
"Deduce whether the material could be X"	16/18	6
Measuring a time period / oscillations accurately	15/18	2
Micrometer or Vernier calipers: choice and use	12/18	1
Mean and uncertainty of a set of repeats	12/18	3
Log-log graph to find a power n; constant from a gradient or intercept	10/18	4
Data logger / light gate as an improvement; assess a suggestion	9/18	6
ln graph for exponential decay	8/18	4
Devise a plan (6 marks)	8/18	5
Safety precaution or hazard; oscilloscope time base	7/18	2, 1

THE SINGLE MOST VALUABLE HABIT

Nearly every extended answer is marked as a **list of independent points** — "an answer that makes reference to **four of the following**". **Write in short, separate, numbered sentences, one idea each.** A flowing paragraph hides points; a list of five crisp statements collects them.

MARKING CONVENTIONS THAT DECIDE YOUR GRADE

e.c.f. is allowed on almost every multi-step calculation — 17 of the 18 mark schemes state it. A wrong early value still earns the later marks if the method is right. **Never leave a calculation blank.**
"Show that" requires every substitution **and** your own answer to **more significant figures** than the given value ("about 200 Hz" → write 227 Hz).
Significant figures are specifically marked on gradients, means and percentage uncertainties — usually 2 or 3 s.f.; give a percentage uncertainty to **1 or 2 s.f.**
Units are required on final answers — but a **log-log gradient has no unit.**
Conclusion marks are dependent: only awarded if the comparison before them is shown. **Always show the number you are comparing with.**

SHEET 1 — INSTRUMENTS, RESOLUTION & SCALES

Instrument	Resolution	Use it for
Metre rule	1 mm	Lengths above about 10 cm
Vernier calipers	0.1 mm (0.01 cm)	Diameters ~1-15 cm; internal diameters and depths
Micrometer screw gauge	0.01 mm	Thicknesses and diameters below ~25 mm — wire, lens edge, rod
Digital calipers	0.01 mm	As Vernier, easier to read
Stopwatch (hand)	0.01 s, but ≈0.1-0.2 s in practice from reaction time	Timing oscillations — always time many and divide
Balance	0.1 g or 0.01 g	Quote the resolution actually shown on the display

RESOLUTION — THE DEFINITION

The **smallest change in the quantity that the instrument can show** — the smallest division on the scale or the last digit on the display. **It is not the uncertainty**: the uncertainty in a single reading is normally **half the resolution**.

CHOOSING AN INSTRUMENT — CALCULATE, DON'T ASSERT

A bare name scores nothing. The scheme wants the **resolution compared with the size of the quantity**, done as a **percentage uncertainty calculation**.

Model answer (d ≈ 2 mm): "A micrometer screw gauge. Its resolution is 0.01 mm compared with 1 mm for a metre rule. For a 2 mm diameter this gives a percentage uncertainty of $(0.005 \div 2) \times 100 = 0.25\%$, compared with $(0.5 \div 2) \times 100 = 25\%$ for a metre rule, so the micrometer gives a far smaller percentage uncertainty."

When the question says "include calculations", **two of the four marks are for the two calculations**. A **larger** quantity measured with the same instrument also has a smaller percentage uncertainty — which is why measuring string **around** a container beats measuring its diameter.

Oscilloscope → frequency: **1.** count the divisions for a **whole number of complete waves** — **use at least two, not one**, to reduce uncertainty · **2.** multiply by the time-base setting, then divide by the number of waves to get $T \cdot 3$. $f = 1/T$.

Worked: two waves in 4.4 divisions at 2.0 ms/div → $2T = 8.8 \times 10^{-3} \text{ s} \rightarrow T = 4.4 \times 10^{-3} \text{ s} \rightarrow f = 227 \text{ Hz} \approx 230 \text{ Hz}$.

Choosing the time base: calculate T, double it, and pick the setting for which **2T fits across the screen while filling as much of it as possible** — the largest trace gives the smallest percentage uncertainty.

Reading a scale: **check and correct for zero error** before using calipers or a micrometer (named in **16 of 18** mark schemes) · read **at eye level, perpendicular** to avoid parallax · on a micrometer **tighten with the ratchet** so the same pressure is applied · record to the **full resolution including trailing zeros** (11.30 mm, not 11.3 mm).

SHEET 2 — TECHNIQUES & SAFETY (17 of 18 papers)

Each point must be a **technique plus its purpose** when the command is "explain".

Technique (any length measurement)	Why
Check and correct for zero error	Removes a systematic error
View the scale at eye level, perpendicular	Avoids parallax error
Measure at several points or orientations, take a mean	The object may not be uniform; reduces the effect of random error
Use a set square against the rule or bench	Ensures the measurement is vertical or horizontal
Measure between the centres of markers / use a fiducial marker	Removes ambiguity about where the measurement starts
Measure a large multiple (e.g. 10 gaps) and divide	Reduces the percentage uncertainty

TIMING OSCILLATIONS — THE MOST REPEATED ITEM ON THE PAPER (3 marks)

1. Use a **timing (fiducial) marker at the centre of the oscillation**, where the object moves fastest, so the moment of passing is easiest to judge.

2. Time a large number of complete oscillations (10 or 20) and divide, which reduces the percentage uncertainty caused by **reaction time**.

3. Repeat and calculate a mean, to reduce the effect of random error.

Also credited: start timing after several oscillations so the motion has settled; count "zero" on the first pass.

"Repeat and take a mean" on its own is capped at one mark. "Use a stopwatch" earns nothing. **Never write that repeating eliminates or removes random error — it only reduces its effect.**

Problem	Technique that scores
Reaction time on a stopwatch	Time many oscillations and divide; or use a light gate and data logger ; or slow the motion (make a ramp less steep)
Heat loss to the surroundings	Insulate or lag; use a lid; start below and finish above room temperature
Judging a maximum or end point	Approach from both directions and average ; use a larger amplitude or longer trace
Background radiation	Measure the background count rate and subtract it from every reading

Hazard	Precautions credited
Radioactive source	Handle with tongs , not fingers; keep as far away as possible; point it away from people; use for as short a time as possible; store in a lead-lined box
Hot liquid / steam / heater	Tongs or heatproof gloves; eye protection; heatproof mat; allow to cool before moving
High voltage / charged capacitor	Discharge the capacitor through a resistor before touching; switch off before rewiring; do not exceed the working voltage
Falling masses	Tray or sand below; keep feet clear; counterweight or clamp the stand
Laser	Do not look into the beam; avoid reflections from shiny surfaces

Precautions must be specific to this experiment. "Wear goggles" and "be careful" score nothing on their own; **"wear eye protection because the spring may snap"** does.

Control variables: name each quantity — "the same mass of liquid, the same initial temperature, the same power supplied, the same beaker" — never "keep everything else the same".

SHEET 3 — UNCERTAINTIES (18/18 papers — the most mechanical marks available)

Situation	Absolute uncertainty
A single reading from a scale or display	Half the resolution. Metre rule $\rightarrow \pm 0.5 \text{ mm}$ · micrometer $\rightarrow \pm 0.005 \text{ mm}$ · 5 cm ³ cylinder $\rightarrow \pm 2.5 \text{ cm}^3$
A length needing two readings (both ends)	The full resolution (two half-resolutions added) — this is why $\Delta h = 65 \text{ mm} \pm 1 \text{ mm}$
A set of repeated readings	Half the range: (largest – smallest) $\div$ 2. Quote mean $\pm$ half range
Given in the question	Use the value given

THE FOUR COMBINING RULES

Adding or subtracting ($a + b$, $a - b$) $\rightarrow$ ADD THE ABSOLUTE uncertainties

Multiplying or dividing (ab , a/b) $\rightarrow$ ADD THE PERCENTAGE uncertainties

Raising to a power (a^n) $\rightarrow$ MULTIPLY the percentage by n (d^4 with 0.9% $\rightarrow$ 3.6%; $\sqrt{a}$ with 4% $\rightarrow$ 2%)

Multiplying by a constant $\rightarrow$ percentage uncertainty unchanged ($2\pi r$ has the same % as r)

Subtraction is the dangerous one: the absolute uncertainty stays the same size while the **value gets small**, so the percentage uncertainty becomes **large**. This is exactly why questions subtract two similar lengths ($t - x$, Δh , $p - q$) — **always convert back to a percentage after a subtraction.**

COMMON ERRORS THAT SCORE ZERO

Using the **full range** instead of **half the range** — called out explicitly in the mark schemes.

Not rounding the uncertainty to **1 (or at most 2) significant figures**, and not quoting the mean to the **same decimal place** as the uncertainty.

Worked: 2.35, 2.37, 2.34, 2.35, 2.33 $\rightarrow$ mean 2.348 = **2.35 mm**; half range $(2.37 - 2.33)/2 = 0.02 \text{ mm}$ $\rightarrow$ **$d = 2.35 \pm 0.02 \text{ mm}$** .

WORKING THROUGH A FORMULA — METHOD MARKS ARE INDEPENDENT

$E = 4MgL^3/wt^3y$, "show that the percentage uncertainty in E is about 3%":

%L = $(0.001 \div 0.798) \times 100 = 0.13\% \rightarrow \times 3 = 0.38\%$

%w: mean 25.02 mm, half range 0.05 mm $\rightarrow 0.20\%$

%t = $(0.02 \div 6.17) \times 100 = 0.32\% \rightarrow \times 3 = 0.97\%$

%y = $(1 \div 85) \times 100 = 1.18\%$

Total = $0.38 + 0.20 + 0.97 + 1.18 = 2.7\% \approx 3\%$

One mark for each percentage and one for the total — write every term on its own line, even if you cannot finish.

Compound area or volume by subtraction: find each term's percentage uncertainty ($\times 2$ for a **squared term**), convert each to an **absolute** uncertainty, then **add the absolutes**. Alternatively compute the **maximum and minimum** possible values and halve their difference — both accepted, but be consistent and show the substitutions.

SHEET 4 — GRAPHS (Q3 every session — the most reliably scored marks on the paper)

WHY A LOG GRAPH — 2 MARKS, BOTH FOR THE ALGEBRA

For $y = kx^n$: $\log y = \log k + n \log x$

"This is in the form $Y = c + mX$, so a graph of $\log y$ against $\log x$ is a **straight line of gradient n and intercept $\log k$** ."

For $C = C_0 e^{-\lambda t}$: $\ln C = \ln C_0 - \lambda t \rightarrow$ "a graph of $\ln C$ against t is a straight line of **gradient $-\lambda$ and intercept $\ln C_0$** ."

The second mark is dependent on writing the log equation first — never skip it.

Mark	What the examiner checks
1-2	Processed data: every log or ln value correct, all to the same number of decimal places (2 d.p. for logs; 3 accepted). One mark per column
3	Axes labelled with the quantity, in the form log (r / fm) where the raw quantity has a unit. A log value itself has no unit
4	Scales: plots must fill more than half the grid in both directions , using sensible divisions (1, 2 or 5 per square). Do not force the origin on
5	Plots accurate to within half a small square (± 1 mm) , as small crosses or encircled dots
6	Best-fit line: a single thin straight line with an even spread of points on both sides . Not dot to dot

SCALES ARE THE MARK MOST OFTEN LOST

Choose the scale AFTER you have computed your log values. Find the smallest and largest, then pick a start value and a step that spread them across the grid. **Logs are often negative — allow for that before you start plotting.**

Taking the gradient: 1. draw the best-fit line **first**; take coordinates **from the line, not from the data table** · 2. use a **large triangle, at least half the length of the line, and mark it on the graph** — the scheme says "correct calculation using large triangle shown" · 3. gradient = $\Delta y / \Delta x$, to 2 or 3 s.f., **with no unit for a log-log graph** · 4. for an intercept, read where the line crosses $x = 0$ **only if $x = 0$ is actually on your grid**, otherwise substitute a point from the line into $c = Y - mX$.

FROM GRADIENT AND INTERCEPT TO THE CONSTANTS

A log-log graph of $\log r$ against $\log A$ gives gradient 0.32 and intercept 0.09:

$n = \text{gradient} = 0.32$ · $\log r_0 = 0.09$, so $r_0 = 10^{0.09} = 1.2 \text{ fm}$ · relationship $r = 1.2 A^{0.32} \text{ fm}$

To get a constant k you must take the ANTILOG of the intercept (10^c , or e^c for an ln graph). **Reading the intercept and calling it k loses the mark.**

Half-life: gradient of $\ln C$ against $t = -\lambda$, then $t_{1/2} = \ln 2 / \lambda = 0.693 / \lambda$. With two isotopes present, the short-lived one decays away first — **use only the later, straight part of the graph.**

SHEET 5 — DEVISE A PLAN (6 marks, 8 of the last 18 papers)

The mark scheme is a list of **six independent points**. Write six numbered sentences and you will collect them.

Point	What to write
1. Independent variable	What you change and how you measure it, naming the instrument : "measure d with a metre rule"
2. Dependent variable	What you measure and with what: "measure the time for 10 oscillations with a stopwatch and divide by 10"
3. Control / set-up detail	One specific precaution for this apparatus: align with a set square; subtract the background count ; keep the mass and initial temperature the same
4. Range and repeats	"Take at least five different values of d , repeating each and taking a mean " — both the number of values and the repeats are credited
5. The graph	State exactly what to plot : "plot T^2 against l ", "plot $\log T$ against $\log M$ ". Derive it from the relationship if a power is involved
6. What the graph shows	State the test: "if the relationship holds the graph will be a straight line through the origin " or "...will have a gradient of -2 "

Proposed relationship	Plot	If it holds
$C = a/d^2$	C against $1/d^2$	Straight line through the origin , gradient a
$P = aL^b$ (b unknown)	log P against log L	Straight line, gradient b , intercept $\log a$
$T = 2\pi\sqrt{m/k}$	T^2 against m	Straight line through the origin, gradient $4\pi^2/k$
$C = C_0 e^{-\lambda t}$	ln C against t	Straight line, gradient $-\lambda$
$T^2 = (\pi^2/g)(2l - h)$	T^2 against h	Straight line, negative gradient $-\pi^2/g$

MODEL PLAN — TESTING $C = a/d^2$ FOR A BETA SOURCE

- Measure the **background count rate with the source removed** and subtract it from every reading to give the corrected count rate.
- Measure the distance d from source to G-M tube with a **metre rule, using a set square**, keeping source and tube aligned.
- Record the count over a **long fixed time (at least one minute)** to reduce the percentage uncertainty caused by the random nature of decay.
- Repeat** the count at each distance and calculate a **mean** corrected count rate C .
- Repeat for **at least five** different values of d .
- Plot C against $1/d^2$; if the prediction is correct the graph will be a straight line through the origin.**

WHERE PLAN MARKS ARE LOST

"Measure the distance" **without the instrument** · "take several readings" **without a number** · "plot a suitable graph" **without saying what against what** · "see if it works" **instead of stating the expected shape**. All score nothing.
Include a labelled circuit diagram whenever electricity is involved — it is usually one of the six marks.

SHEET 6 — EVALUATION & CONCLUSIONS

"DEDUCE WHETHER THE MATERIAL COULD BE X" — THE STANDARD Q4 ENDING (16/18 papers)

Method A — percentage difference against percentage uncertainty

1. % difference = $|\text{experimental} - \text{accepted}| \div \text{accepted} \times 100$ · **2.** compare with the percentage uncertainty you calculated earlier · **3.** conclude: "The percentage difference is **3.9%**, which is **greater than** the percentage uncertainty of **0.96%**, **so the cone is not made from brass.**"

If %difference $\leq$ %uncertainty, the value is **consistent** with the accepted one.

Method B — overlap of ranges

1. Work out value $\pm$ absolute uncertainty · **2.** state whether the accepted value lies inside that range · **3.** conclude: "The lower limit is **8.81 g cm⁻³**, which is greater than **8.53 g cm⁻³**, so the cone is not made from brass."

The accepted value must appear as a number in your sentence — "the values are close so it is brass" scores nothing. **Always divide by the accepted value**, not your experimental one. If a **range** of accepted values is given, test whether your range **overlaps** it.

Term	Definition	How it is judged
Accurate	Close to the true (accepted) value	Compare the mean with the true value
Precise	Repeat readings are close to one another	Compare the range or spread of the repeats

Model answer: "Both students have the same mean, so they have the **same accuracy**; but **without the true value we cannot tell how accurate** either is. Student B's readings have a **smaller range**, so Student B's data are **more precise**."

Precise is not accurate: a zero error or other systematic error shifts every reading by the same amount, so they agree closely with each other but are **all wrong**.

Error	Cause	Effect	Reduce it by
Random	Reaction time, judging an end point, fluctuating readings	Scatters points either side of the line; affects precision	Repeating and taking a mean; timing many oscillations
Systematic	Zero error, wrong calibration, parallax, heat loss, background radiation	Shifts every reading the same way ; affects accuracy ; shows as a wrong intercept	Checking zero error, correcting for background, insulating. Repeating does not help

JUDGING WHETHER RESULTS SUPPORT A RELATIONSHIP — FOUR STEPS

1. State what the graph **actually shows**: "the points lie on a straight line" / "the gradient is 0.51" · **2.** state what the relationship **predicts**: " $T \propto \sqrt{M}$ requires a gradient of 0.5" · **3.** say whether they agree **within experimental uncertainty** · **4.** give the verdict: "so the results support the suggested relationship".

If asked about validity, check the intercept too: a straight line that does **not pass through the origin** when the relationship predicts it should indicates a **systematic error**, so the relationship as stated is not supported.

Why results may not give an accurate value (recurring 4-marker): too **few data points** to draw a reliable best-fit line · the **range is too small** · there may be a **systematic error** in one measured quantity (say which, and why) · **so an accurate best-fit line cannot be drawn and the gradient may not be close to the true value**. The scheme explicitly **ignores** comments about significant figures, repeating readings and uneven intervals here.

Suggested improvement	Benefit	Limitation / verdict
Longer card through a light gate	A longer time is measured, so smaller percentage uncertainty in the time	Must still be shorter than the gap between gates; the average velocity over a longer card is less representative of the instant of collision
Data logger with a light gate	Removes reaction-time error ; many readings quickly; two quantities simultaneously	Only worth it if the timing uncertainty dominates ; must be set up and calibrated correctly
Record on video	Replay frame by frame , so the moment of an event is judged more accurately	Resolution limited by the frame rate
Make a ramp less steep	Slower motion → longer time → smaller percentage uncertainty ; finish line easier to judge	Too shallow and the object may not roll freely

SHEET 7 — DO NOT WRITE / WRITE INSTEAD

Do not write	Write instead
"Repeat and take a mean" as a whole 3-mark answer	One point only — add a timing marker, timing many oscillations, and starting after the motion settles
"Eliminates / removes / avoids random error"	Reduces the effect of random error
"Use a stopwatch" / "measure carefully" / "be accurate"	A named technique with its purpose
"Wear goggles" / "be careful" with no context	A precaution matched to the stated hazard
"Keep everything else the same"	Name each control variable
"Use a micrometer because it is more accurate"	"Because its resolution is 0.01 mm, giving a percentage uncertainty of 0.25% rather than 25% "
Full range as the uncertainty of repeats	Half the range: $(\text{max} - \text{min}) \div 2$
Resolution as the uncertainty of a single reading	Half the resolution — but the full resolution where two ends are read
Adding absolute uncertainties when multiplying	Add percentages for $\times$ and $\div$; multiply by the power for a^n
Taking the intercept of a log graph as k	$k = 10^{\text{intercept}}$, or $e^{\text{intercept}}$ for an $\ln$ graph
A unit on a log-log gradient	No unit
Reading gradient coordinates from the data table	Read them from the best-fit line , using a large triangle drawn on the graph
Joining the plotted points dot to dot	A single thin best-fit line with an even spread of points either side
Log values to mixed decimal places	All values in a column to the same number of decimal places (2 d.p.)
"Plot a suitable graph"	Plot y against x , naming both , and state the expected shape
"Take several readings"	At least five (or six) values over as wide a range as possible, each repeated
"The values are close, so it is steel"	A stated percentage difference compared with a stated percentage uncertainty , or limits compared with the accepted value
"It is precise so it is accurate"	Precision is the spread of repeats ; accuracy needs comparison with the true value
"Human error"	The specific error — reaction time, parallax, zero error, heat loss

SHEET 8 — FINAL CHECKLIST

- ☐ I know the **resolution of every instrument** and can justify a choice with **two percentage-uncertainty calculations**.
- ☐ I can write the **three-point answer** for measuring a time period accurately, and a technique **with its purpose** for any length measurement.
- ☐ I can give **two specific safety precautions** for a radioactive source, a hot liquid, a charged capacitor and falling masses.
- ☐ I use **half the resolution** for a single reading, the **full resolution** when two ends are read, and **half the range** for repeats.
- ☐ I **add absolutes** for $+$ and $-$, **add percentages** for $\times$ and $\div$, and **multiply by the power** for a^n .
- ☐ I can **take logs of a relationship**, state the $Y = c + mX$ form, and say what the gradient and intercept represent.
- ☐ I can score **all six plotting marks**: processed data to 2 d.p., labelled axes, scales filling over half the grid, plots to ± 1 mm, single best-fit line.
- ☐ I take gradients **from the line with a large triangle drawn on the graph**, give **no unit** for a log-log gradient, and take an **antilog** to get k .
- ☐ I can write a **six-point plan** naming instruments, a control, at least five values with repeats, the graph, and the expected shape.
- ☐ I finish every "deduce" by **comparing a stated percentage difference with a stated percentage uncertainty, naming the accepted value**.
- ☐ I can distinguish **accuracy from precision**, and **random from systematic** error, and say which can be reduced by repeating.
- ☐ I avoid **every phrase on sheet 7**.

WALKING INTO THE EXAM

A mark is worth **1.5 minutes** — you are not short of time. **Leave 6-8 minutes for the Q3 graph and do not rush the axes**; it is worth roughly a third of the paper and the six plotting marks are the most reliably scored on the whole paper. Write every extended answer as **short numbered sentences, one idea each**. Never leave a calculation blank — **e.c.f. is allowed almost everywhere**. And end every "deduce" with **two numbers and a verdict**.